

SEE 2083: Compulsory Mathematics

Complete Formula Sheet

1. Sets

Cardinality of Two Sets (A and B):

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n_0(A) = n(A) - n(A \cap B)$
- $n(U) = n(A \cup B) + n(\overline{A \cup B})$

2. Arithmetic (Tax & Finance)

Discount and VAT:

- Discount Amount = Marked Price (MP) $\times \frac{\text{Discount \%}}{100}$
- Selling Price (SP) = MP - Discount Amount
- VAT Amount = SP $\times \frac{\text{VAT \%}}{100}$
- SP with VAT = SP + VAT Amount

Compound Interest (Annually):

- Amount (CA) = $P \left(1 + \frac{R}{100}\right)^T$
- Compound Interest (CI) = $P \left[\left(1 + \frac{R}{100}\right)^T - 1 \right]$

Population Growth: $P_T = P_0 \left(1 + \frac{R}{100}\right)^T$

3. Mensuration (Area & Volume)

Triangles:

- Right Angled / General: $A = \frac{1}{2} \times \text{base} \times \text{height}$
- Scalene (Heron's Formula): $A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$
- Equilateral: $A = \frac{\sqrt{3}}{4}a^2$
- With given angle: $A = \frac{1}{2}ab \sin \theta$

Solids (Cylinder, Sphere, Cone):

- **Cylinder:** CSA = $2\pi rh$, TSA = $2\pi r(r+h)$, Volume = $\pi r^2 h$
- **Sphere:** Surface Area = $4\pi r^2$, Volume = $\frac{4}{3}\pi r^3$
- **Hemisphere:** CSA = $2\pi r^2$, TSA = $3\pi r^2$, Volume = $\frac{2}{3}\pi r^3$
- **Cone:** CSA = πrl , TSA = $\pi r(r+l)$, Volume = $\frac{1}{3}\pi r^2 h$ (where $l^2 = r^2 + h^2$)

4. Algebra

Indices:

- $a^m \times a^n = a^{m+n}$
- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $a^{-n} = \frac{1}{a^n}$
- $a^0 = 1$

Quadratic Equation ($ax^2 + bx + c = 0$): $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

5. Statistics

For Continuous (Grouped) Data:

- **Mean:** $\bar{x} = \frac{\sum fm}{N}$ (where m is the mid-value of the class)
- **Median Class:** Value of $\left(\frac{N}{2}\right)^{th}$ item
- **Median (M_d):** $L + \frac{\frac{N}{2} - cf}{f} \times i$
- **First Quartile (Q_1):** $L + \frac{\frac{N}{4} - cf}{f} \times i$
- **Third Quartile (Q_3):** $L + \frac{\frac{3N}{4} - cf}{f} \times i$

(Where L = lower limit, cf = cumulative frequency of preceding class, f = frequency, i = class size)

6. Probability

Basic Rules:

- $P(E) = \frac{n(E)}{n(S)}$
- $P(\text{not } E) = 1 - P(E)$
- Mutually Exclusive Events: $P(A \cup B) = P(A) + P(B)$
- Independent Events: $P(A \cap B) = P(A) \times P(B)$